Author's Note

CSYE 7270 — Take-Home Midterm

Page 1 — Design Choices

I chose Category E because it offered the clearest path to a mechanistically demonstrable failure case. Most Category A and B topics produce failures that are aesthetic or subjective — "the AI-generated art looks off" or "the NPC behavior feels wrong." Those failures are real, but they are hard to trigger precisely and harder to reproduce. Context window truncation is different: it is a technical failure with an exact causal chain, triggerable by changing a single integer in a line of code. That reproducibility was the deciding factor.

The specific topic — runtime narrative branching and context management — also felt genuinely underexplored. Most public writing about LLM-driven NPCs stays at the feature layer: "imagine an NPC that remembers everything." Almost none of it addresses the architectural question of how memory is actually managed when the history grows longer than the context window. That gap is where this essay lives.

I also considered Category B — behavior trees vs. LLM-driven NPC dialogue — but rejected it because the failure mode there is harder to isolate. The comparison between two systems introduces too many variables. Category E allowed me to focus on a single architectural decision and its consequences, which is a cleaner instance of the master claim.

What I left out was intentional. I did not cover token-level implementation details, cost optimization, or latency — all real concerns in a shipped game. I also did not address the case where the LLM itself hallucinates events that never happened, which is a separate failure mode from the one this essay addresses. These omissions were made to keep the argument tight: one failure mode, one causal chain, one correction.

The essay is a specific instance of the course's master claim — "AI is a pipeline decision, not a magic layer" — in the following way: the NPC's character is not determined by the LLM's capability. It is determined by what history you pass into the prompt. The pipeline decision — what to keep, what to discard, how to compress — precedes the model call. Change the pipeline decision and you change the character, without changing the model at all. That is the claim made concrete.

This is also why the topic sits in Category E rather than Category B. A behavior tree comparison would have demonstrated AI's role in game mechanics. This essay demonstrates AI's role in game identity — what a character is allowed to know, and therefore what kind of character it can be. That distinction felt worth arguing for on its own terms.

Page 2 — Tool Usage

Bookie the Bookmaker was used to draft prose bridging intuition to mechanism. The initial draft Claude generated for The Mechanism section read:

> *"This history is passed as a list of message objects: alternating user and assistant turns that reconstruct the conversation."*

I kept this line but restructured the paragraph order. Claude's draft placed the technical implementation before the intuitive concept — explaining message objects before explaining why memory fails. Eddy flagged this as a sequencing problem: "The reader needs to feel the problem before they need to understand the plumbing." I moved the core fact — "an LLM has no persistent memory. Every call starts from zero." — to the top of the section, before any technical detail. This was a substantive correction: the argument only works if the reader understands the constraint before the mechanism.

Eddy the Editor audited the full draft and flagged two structural problems. First, the thesis sentence was buried in the third paragraph of The Scenario — Eddy noted it was "doing the heavy lifting of a transition but currently positioned as a throwaway." I moved it to the end of The Scenario as the section's closing argument. Second, the Design Decision section read as a taxonomy rather than an argument: "The reader needs to understand not just what the three options are but what kind of game designer chooses each one and why." I revised each strategy to include a specific use case and failure consequence. I rejected Eddy's suggestions about Substack SEO and publish timing — these were not relevant to a technical essay submission.

Figure Architect was used to generate figure prompts for three high-assertion claims:

- Figure 1: A timeline diagram showing how truncation deletes high-salience early events first
- Figure 2: A comparison table of the three context strategies across four dimensions
- Figure 3: Three side-by-side NPC response panels showing what is retained and lost under each condition

These figure prompts appear in the essay body as descriptive placeholders that could be executed in Midjourney or a similar tool. The decision to include prompts rather than rendered images reflects a prioritization of argumentative completeness over visual polish. Each prompt was written to make the abstract mechanism visible — the timeline diagram, the strategy comparison table, and the side-by-side NPC response panels each anchor a claim that would otherwise require the reader to hold multiple variables in mind simultaneously.

Page 3 — Self-Assessment

Argumentative Rigor (35 pts) — estimated: 32/35

The essay makes a specific claim: context management is a narrative architecture decision, and the wrong strategy produces a specific, observable failure. The failure case is mechanistically explained — truncation by recency → loss of high-salience early events → NPC response inconsistent with game history → immersion breaks. The failure is triggered and observed in the demo with real API outputs, not described in prose. The deduction is for the Figure descriptions, which are prompts rather than rendered images, and for the Design Decision section which could go deeper on salience-weighted retention.

Technical Implementation (25 pts) — estimated: 23/25

The demo handles real API calls with real outputs. The Mandatory Human Decision Node is present, specific, and documented in the required format. The failure case is reproducible by changing a single variable. The deduction is for the compression strategy being relatively simple — a more sophisticated implementation would use LLM-generated summaries rather than concatenated assistant responses.

Clarity (20 pts) — estimated: 20/20

The essay follows the five-section structure. Every claim has a mechanism. The Figure descriptions serve as visual anchors for the key arguments.

Total estimated: 75/80 core competency points

The failure case this demo demonstrates is clean: a reader can clone the repo, change `full_history[-2:]` to `full_history[-1:]`, run the notebook, and observe Victor responding as if the civilian killings never happened. What it does not demonstrate is the failure mode at scale — a real game session with hundreds of exchanges, where the compression strategy itself begins to degrade. That is the honest limit of this proof-of-concept.